

Shayan Shakeri

(416) 471-6651 | shayansnezhad@gmail.com | github.com/sshakerinezhad | merlyn-labs.com | sshakerinezhad.github.io |
Toronto, Canada

PROFESSIONAL SUMMARY

AI Research Engineer focused on RL environments and hard-to-game evaluation for frontier models. Built automated graders that resist reward hacking: VLM judges that score agent rollouts into dense, context-dependent rewards, and a deterministic agent eval pipeline that surfaced systematic risk-downplaying bias in a \$200B+ AUM GenAI tool, and authored an end-to-end RL training environment integration (flow-matching VLA for RLinf). Co-founder of Merlyn Labs; M.Eng in AI & Robotics, University of Toronto, expected April 2027.

CORE COMPETENCIES

- RL Environments
- Automated Graders
- Reward Modeling
- Hard-to-Game Evals
- Agent Evaluation
- Failure Analysis

WORK EXPERIENCE

Merlyn Labs Aug 2025 - Present

AI Researcher & Co-Founder | Toronto, Canada

- Building VLM judges (automated graders) that score agent rollouts into dense, context-dependent RL rewards designed to resist gaming.
- Authored an end-to-end RL training environment: open-sourced a flow-matching VLA integration for RLinf, enabling RL on the BEHAVIOR-1K suite in OmniGibson.
- Ran failure analysis on frontier VLA models: showed pi-0.5 "generalization" on LIBERO-PRO is recipe-induced overfitting; conservative finetuning doubled position-swap success (21% to 42%).

BMO AI Centre of Excellence Sep 2024 - Present

AI Research Engineer | Toronto, Canada

- Uncovered systematic bias to downplay investment risk in a GenAI tool serving \$200B+ AUM wealth management.
- Built a deterministic agent eval pipeline on hundreds of synthesized inputs to detect misaligned model outputs at scale.
- Building a graph-based agentic system answering complex multi-hop relational queries across bank client data.

Epineuron Technologies May 2021 - Aug 2022

Biomedical & Electrical Engineering Co-op | Mississauga, Canada

- Helped develop PeriPulse (FDA Breakthrough-designated neurostimulation device, now in multi-national trials); designed and assembled PCBs and authored IEC 60601-1 / ISO 13485 validation protocols.

PROJECTS

BardSong: AI Storytelling Pipeline

Built an end-to-end software pipeline: Groq speech-to-text to Gemini scene extraction to image generation to narrated video, with an LLM fine-tuned on a domain corpus. Currently in closed alpha with 23 users.

Python, End-to-End Pipeline, LLM Fine-tuning, Multi-Stage Orchestration

Stanford BEHAVIOR-1K Challenge 8th Place, Standard Track

Rigorous failure analysis on frontier policies: identified proprioceptive collapse (60% masking improved task success up to 48%) and found chunked execution outperformed temporal ensembling 3x. Trained on 10,000+ demonstrations across 22 tasks; published a technical report.

Reinforcement Learning, Failure Analysis, Benchmark Rigor, PyTorch

EDUCATION

M.Eng in AI & Robotics, University of Toronto Sep 2024 - Apr 2027 (anticipated)

Coursework: Deep Learning, Reinforcement Learning, AI Applications in Robotics, Healthcare Robotics.

B.Eng, Engineering Physics Co-op, McMaster University Sep 2018 - Jun 2023

Dean's Honour List. Coursework: Computational Multiphysics, Quantum Computing, Numerical Methods, Signals & Systems.

SKILLS

Evaluation & RL Environments: RL Environments, Automated Graders, Reward Modeling, Hard-to-Game Evals, Agent Evaluation, Reward-Hacking Resistance, Failure Analysis, Benchmark Design

ML, Agents & Languages: LLM Fine-tuning, Reinforcement Learning, Imitation Learning, VLM Judges, Multi-Agent Systems, Alignment Evaluation, Python, C/C++, PyTorch, JAX